Brain, Behavior, and Immunity 28 (2013) 159–169

Contents lists available at SciVerse ScienceDirect

Brain, Behavior, and Immunity

journal homepage: www.elsevier.com/locate/ybrbi

Altered immune pathway activity under exercise challenge in Gulf War Illness: An exploratory analysis

Gordon Broderick [a,*,1], Rotem Ben-Hamo [b,2], Saurabh Vashishtha [a,3], Sol Efroni [b,4], Lubov Nathanson [c,5], Zachary Barnes [d,6], Mary Ann Fletcher [d,7], Nancy Klimas [c,e,8]

[a] Department of Medicine, University of Alberta, Edmonton, Canada
[b] Goodman Faculty of Life Science, Bar Ilan University, Ramat Gan, Israel
[c] Miami Veterans Affairs Medical Center, Miami, FL, USA
[d] Department of Medicine, University of Miami, Miami, FL, USA
[e] Institute for Neuro-Immune Medicine, Nova Southeastern University, Ft. Lauderdale, FL, USA

ARTICLE INFO

Article history:
Received 5 July 2012
Received in revised form 28 October 2012
Accepted 13 November 2012
Available online 29 November 2012

Keywords:
Gene expression
Pathway activity
Gulf War Illness
Exercise challenge
Mathematical immunology

ABSTRACT

Though potentially linked to the basic physiology of stress response we still have no clear understanding of Gulf War Illness (GWI), a debilitating illness presenting with a complex constellation of immune, endocrine and neurological symptoms. Here we compared male GWI ($n = 20$) with healthy veterans ($n = 22$) and subjects with chronic fatigue syndrome/myalgic encephalomyelitis (CFS/ME) ($n = 7$). Blood was drawn during a Graded eXercise Test (GXT) prior to exercise, at peak effort (VO2 max) and 4-h post exercise. Affymetrix HG U133 plus 2.0 microarray gene expression profiling in peripheral blood mononuclear cells (PBMCs) was used to estimate activation of over 500 documented pathways. This was cast against ELISA-based measurement of 16 cytokines in plasma and flow cytometric assessment of lymphocyte populations and cytotoxicity. A 2-way ANOVA corrected for multiple comparisons (q statistic <0.05) indicated significant increases in neuroendocrine-immune signaling and inflammatory activity in GWI, with decreased apoptotic signaling. Conversely, cell cycle progression and immune signaling were broadly subdued in CFS. Partial correlation networks linking pathways with symptom severity via changes in immune cell abundance, function and signaling were constructed. Central to these were changes in IL-10 and CD2+ cell abundance and their link to two pathway clusters. The first consisted of pathways supporting neuronal development and migration whereas the second was related to androgen-mediated activation of NF-κB. These exploratory results suggest an over-expression of known exercise response mechanisms as well as illness-specific changes that may involve an overlapping stress-potentiated neuro-inflammatory response.

© 2012 Elsevier Inc. All rights reserved.

1. Introduction

Within months of returning from Operation Desert Storm an alarming number of Gulf War veterans began reporting fatigue, musculoskeletal discomfort, skin rashes, and cognitive dysfunction (Haley, 1997; Fukuda et al., 1998; Wolfe et al., 1998). We still have no clear understanding of Gulf War Illness (GWI) although the basic physiology of response to stress whether psychological, chemical or other provides a starting point. Indeed clinical presentation of GWI overlaps strongly with that of another stress-mediated illness: Chronic fatigue syndrome/myalgic encephalomyelitis (CFS/ME) (Kang et al., 2003; Eisen et al., 2005). Though brain imaging has shown promise (Liu et al., 2011; Haley et al., 2009), single biomarkers for these illnesses in the lymphocyte transcriptome remain elusive (Byrnes et al., 2009). Early RT-PCR amplification of cytokine associated transcripts by Zhang et al. (1999) indicated that Gulf War veterans diagnosed with chronic fatigue syndrome (CFS/ME)

* Corresponding author. Address: Div. of Pulmonary Medicine, Department of Medicine, University of Alberta, WMC 2E4.41 WC Mackenzie Health Sciences, 8440-112 Street, Edmonton, AB, Canada T6G 2R7. Tel.: +780 492 1633; fax: +780 407 3027.

E-mail addresses: gordon.broderick@ualberta.ca (G. Broderick), rotem@systemsbiomed.org (R.Ben Hamo), vashisht@ualberta.ca (S. Vashishtha), sefroni@mail.biu.ac.il (S. Efroni), lubov7@yahoo.com (L. Nathanson), ZBarnes@med.miami.edu (Z. Barnes), MFletche@med.miami.edu (M.A. Fletcher), nklimas@nova.edu (N. Klimas).

[1] Tel.: +1 780 492 1633.
[2] Tel.: +1 780 566 9406.
[3] Tel.: +972 3 7384519.
[4] Tel.: +972 3 7384518.
[5] Tel.: +1 305 607 8227.
[6] Tel.: +1 305 243 6218.
[7] Tel.: +1 305 243 6288.
[8] Tel.: +1 305 575 3267.

0889-1591/$ - see front matter © 2012 Elsevier Inc. All rights reserved.
http://dx.doi.org/10.1016/j.bbi.2012.11.007

had significantly higher levels of IL-2, IL-10, IFN-γ, and TNF-α transcript than healthy controls while non-veteran CFS/ME subjects did not differ. More recent work by our group (Whistler et al., 2009) focused on the expression in Gulf War veterans of transcripts associated with natural killer (NK) cell function before, during and after a maximal graded exercise challenge (VO2 max). In this specific group of 108 microarray probe sets, 49 were increased in expression 2-fold and none were decreased in GWI. Specifically, mRNA for perforin (PRF1) and granzyme (GZMB), along with killer cell lectin-like receptors (KLR) and death receptor induction (FASLG) indicated a depressed NK cell cytotoxic response to exercise in GWI.

Additional surveys have been done using animal models and have focused on exposure to chemical agents used in theatre; namely organophosphates and acetylcholine esterase (AChE) inhibitors such as pyridostigmine bromide (PB) (Golomb, 2008; Amourette et al., 2009) and sarin (Shewale et al., 2012; Mach et al., 2008). Though much of this has focused on proteomic and phospho-proteomic profiling (Torres-Altoro et al., 2011; Zhu et al., 2010), recent work by Barbier et al. (2009) surveyed the expression of genes associated with stress response, learning and memory in the hippocampus and hypothalamus of mice. Their results indicated that stressed animals exposed to PB showed increases in hippocampal expression of three genes implicated in memory development: brain-derived neurotrophic factor (BDNF), tropomyosin-related kinase B (TrkB) and calcium/calmodulin-protein kinase II alpha (CamKII). This type of survey is not easily performed in human subjects however circulating lymphocytes have been shown to reflect changes in the brain (Mutez et al., 2011).

As cholinergic signaling is a known modulator of immune response (Kawashima and Fujii, 2003) we wanted to explore if NF-κB activity might be altered in the circulating lymphocytes of GWI subjects perhaps as a lasting result of exposure in theatre to the AChE inhibitors described above. We used a maximal exercise challenge to improve detection as NF-κB (Kim et al., 2009) and many other components of immune function (Walsh et al., 2011) are responsive to exercise as are neuro-endocrine modulators thereof, such as cortisol and neuropeptide Y (NPY) (Jonsdottir, 2000). Consistent with this, our group found previously that IL-1, an NF-κB modulator, exerted a strong influence on exercise-induced immune cell signaling in GWI (Broderick et al., 2011). Based on this, we hypothesized that cholinergic modulation of endocrine-immune function during exercise would differ in GWI subjects. This would manifest not only through changes in a variety of inflammatory pathways but also in associated metabolic co-regulators such as mTOR (Waickman and Powell, 2012; Chi, 2012) as well as in neurotransmission pathways responsive to exercise (Jonsdottir, 2000).

To evaluate these effects we performed an exploratory analysis of gene expression using a novel methodology (Efroni et al. (2007, 2008)) for estimating the activity level of over 400 pathways described in the National Cancer Institute (NCI)/Nature Pathway Interaction Database (PID) (Schaefer et al., 2008) and the Kyoto Encyclopedia of Genes and Genomes (KEGG) database (Kanehisa et al., 2010). Subjects were then compared in terms of the activity level of these pathways. This is a significant departure from conventional analysis, which is based only on those pathways supported by genes expressed significantly across groups (Kerr et al. 2008a,b; Whistler et al., 2009) and where no estimate of pathway activation is produced. Using this alternative approach we estimated pathway activation from gene expression in peripheral blood mononuclear cells (PBMC) collected before exercise, at peak effort and 4 h after exercise in veterans with GWI and healthy veterans. We also used a smaller pilot cohort of CFS/ME subjects as a disease control group. Using a 2-way ANOVA and correcting for false discovery we found 19 pathways uniquely expressed in GWI compared to CFS/ME and healthy control subjects. To further enforce biological relevance we constructed association networks describing the cascade of pathway activation to lymphocyte population, cytotoxic function, cytokine signaling and symptom severity in GWI. These suggested a broad support of pathways involved in neuronal development and migration as well as hormone-mediated NF-κB activation and changes in apoptotic signaling. While elements of these signatures overlapped in part with expected exercise response mechanisms, we found that their activity was disproportionate in GWI and that components not normally responsive to exercise were being recruited.

2. Material and methods

2.1. Sample collection and processing

2.1.1. Cohort recruitment

As part of a larger ongoing study a subset of CFS/ME ($n = 7$), GWI subjects ($n = 20$) and healthy but sedentary Gulf War era veterans ($n = 22$) were recruited from the Miami Veterans Administration Medical Center. All subjects were comparable in age, body mass index (BMI), ethnicity and duration of illness. Subjects were male and ranged in age between 30 and 55. Inclusion criteria was derived from Fukuda et al. (1998), and consisted in identifying veterans deployed to the theater of operations between August 8, 1990 and July 31, 1991, with one or more symptoms present after 6 months from at least 2 of the following: fatigue; mood and cognitive complaints; and musculoskeletal complaints. Subjects were in good health prior to 1990, and had no current exclusionary diagnoses (Reeves et al., 2003). Medications that could have impacted immune function were excluded. Use of the Fukuda definition in GWI is supported by Collins et al. (2002). Summary results of subject demographics and exercise performance are listed in Table 1.

Ethics statement. All subjects signed an informed consent approved by the Institutional Review Board of the University of Miami. Ethics review and approval for data analysis was obtained by the IRB of the University of Alberta.

2.1.2. Subject assessment

All subjects received a physical examination and medical history including the GWI symptom checklist as per the case definition. Psychometric questionnaires included the Multidimensional Fatigue Inventory (MFI) (Smets et al., 1995), a 20-item self-report instrument designed to measure fatigue, and the Medical Outcomes Study 36-item short-form survey (SF-36) (Ware and Sherbourne, 1992) assessing health-related quality of life. The Krupp Fatigue Severity Inventory (Krupp FSI) was also used to measure perceptions of fatigue severity (Krupp et al., 1989) while the impact of symptoms on the activities of daily life was measured with the Sickness Impact Profile (SIP) (Bergner et al., 1981). Subjects

Table 1
Summary of mean and standard error values () for demographic variables and exercise performance in GWI, CFS/ME and healthy control groups.

	HC	CFS	GWI
Number of subjects	22	7	20
Race			
Caucasian	4	6	7
Hispanic	11	1	9
African American	6	0	4
Asian	1	0	0
Age (years)	41.2 (1.4)	45.7 (2.3)	42.7 (1.4)
Body–mass index (BMI) (kg/m^2)	27.9 (0.9)	26.2 (1.3)	28.3 (0.7)
Peak VO2 max (mL/kg/min)	27.6 (1.1)	21.7 (0.9)[a]	24.9 (1.0)
% Predicted peak VO2 max (%)	76.0 (3.2)	63.7 (2.9)[a]	67.9 (2.4)
Time to VO2 max (min)	8.1 (0.6)	6.3 (0.8)	7.2 (0.4)

[a] Two-tailed t test significant at 0.05 level versus HC.

were screened for quantity and quality of sleep, and evaluated for the likelihood of primary sleep disorders using the Pittsburgh Sleep Quality Index (PSQI) (Buysse et al., 1989). Designed to assess symptoms of post-traumatic stress disorder (PTSD), the Davidson Trauma Scale (DTS) (Davidson et al., 1997) was applied to those subjects who reported a traumatic experience (death of loved one, assault, injury, etc.…). This instrument is divided into three components: intrusion, avoidance, and hyper-arousal. While 66% of the GWI cohort presented DTS scores consistent with PTSD, this data is available only for a small fraction of CFS/ME subjects and healthy control subjects. Finally, aspects of cognitive impairment were assessed using the Paced Auditory Serial Addition Task (PASAT). This serial-addition task is used to assess the rate of information processing, sustained attention, and working memory (Gronwall, 1977). A summary of symptom severity measures and results obtained in each group is presented in Supplemental table S1.

Immune response was stimulated with a standard maximal Graded eXercise Test (GXT) using a Vmax Spectra 29c Cardiopulmonary Exercise Testing Instrument, Sensor-Medics Ergoline 800 fully automated cycle ergometer, and SensorMedics Marquette MAX 1 Sress ECG. According to the McArdle protocol (McArdle et al., 2007) subjects pedaled at an initial output of 60 W for 2 min, followed by an increase of 30 W every 2 min until the subject reached: (1) a plateau in maximal oxygen consumption (VO2); (2) a respiratory exchange ratio >1.15; or (3) the subject stopped the test. A first blood draw was conducted prior to exercise following a 30-min rest. Second and third blood draws were conducted upon reaching peak effort (VO2 max) and at 4-h post exercise respectively. Summary exercise performance results are presented in Table 1. All control subjects were screened as sedentary based on their response to a questionnaire upon recruitment. This was subsequently confirmed by the aerobic capacities obtained (VO2 max) (Hurwitz et al., 2010). Adjusted maximum VO2 levels were comparable between GWI and control groups but were significantly lower for the CFS disease control group ($p < 0.05$). Trends existed towards shorter exercise bouts in both illness groups but these did not achieve statistical significance.

2.1.3. Gene expression

At each blood draw three 8-mL tubes of blood were collected in CPT vacutainers (B-D-Biosciences, San Jose, CA). The peripheral blood mononuclear cells (PBMC) were isolated and stored in liquid nitrogen under conditions designed to maintain viability. Specifically, whole blood was added to Ficoll-Paque, centrifuged at 1000*g* for 25 min. PBMC's were isolated from the PBMC ring atop the Ficoll layer into a separate tube, centrifuged at 300*g* for 10 min, then re-suspended in PBS. Cells were then counted using a Beckman Coulter viCell, and cryopreserved in freezing media (temperature lowered 1 °C/min until @ −80 °C).

Total RNA was extracted using TRI Reagent (Molecular Research Center, Cincinnati, OH) following the manufacturer's protocol. The quality and quantity of RNA was assessed using the Agilent Bioanalyzer 2100 RNA 6000 Nano Kit (Agilent Technologies, CA). From each sample, 300 ng of total RNA was converted into cDNA by reverse transcription using a T7-oligo(dT) primer and the Affymetrix 3′ IVT Express Kit (Affymetrix, Santa Clara, CA) according to standard manufacturer protocol. The generated cDNA was purified using the GeneChip Sample Cleaning Module (Affymetrix) and labeled cRNA was generated by in vitro transcription using the biotinylated nucleotide mix. This was then purified with the Cleaning Module and quantified using the Nanodrop ND-1000 spectrophotometer (NanoDrop Technologies, Inc., Wilmington, DE USA). In each preparation 11 μg cRNA was fragmented in Fragmentation Buffer (Affymetrix) in a final reaction volume of 25 μl.

Hybridization, washing, staining and scanning were done using Affymetrix GeneChip instruments (Hybridization Oven 640, Fluidics Station 450Dx, Scanner GCS3000Dx) and Affymetrix Human U133 2.0 arrays (Affymetrix) as per manufacturer's standards. Microarray image files (.cel data) were generated using the Affymetrix GCOS software tool with default microarray analysis parameters to provide overall within chip normalization of the image intensity distribution. The quality parameters that were monitored besides cRNA total yield and cRNA A260/A280 ratio included: (i) background noise (*Q* value), (ii) percentage of present called probe sets, (iii) scaling factor, (iv) information about exogenous *Bacillus subtilis* control transcripts from the Affymetrix Poly-A control kit (lys, phe, thr, and dap), and (v) the ratio of intensities of 3′ probes to 5′ probes for a housekeeping gene (GAPDH).

In a quality control step a custom panel of 10 gene transcripts, plus GAPDH as a housekeeping gene, were surveyed using the NanoString nCounter system (http://www.nanostring.com/products/assays/). This code set contained a 3′ biotinylated capture probe and a 5′ reporter probe tagged with a fluorescent barcode, two sequence-specific probes for each of 10 transcripts. Probes were hybridized in triplicate for 19 h at 65 °C to 100 ng of total RNA collected from 15 patients (5 patients with CFS, 5 patients with GWI and 5 healthy control patients) for all three time points. Excess capture and reporter probes were removed and transcript-specific ternary complexes were immobilized on a streptavidin-coated cartridge using the NanoString preparation station robotic fluids handling platform as per manufacturer's standard protocol. The nCounter Digital Analyzer was used to count individual fluorescent barcodes and quantify target RNA molecules present in each sample. Background hybridization signal was determined using spike in negative controls, and all mRNAs with expression below mean background + 2 standard deviations were considered below the detection limit. Raw code count was then normalized based on the relative number of positive control counts. This background was then subtracted from the raw transcript count. A comparison of nCounter and mRNA-seq data is available in Sun et al. (2011). Analysis was conducted at the Oncogenomics Core Facility at the University of Miami (Miami, USA) Results of this partial validation step are presented in Table S2.

2.1.4. Cell cytometry and cytotoxicity

Flow cytometry was performed on each sample to determine lymphocyte subset abundance using a Beckman/Coulter FC500. Whole blood samples were stained in 5 color combinations, with the appropriate concentrations of antibodies, erythrocytes lysed and the cells fixed with the Optilyse C reagent (Beckman-Coulter Corp., Hialeah, FL). Lymphocyte, monocyte and granulocyte populations were determined using light scatter and back gating on fluorescence for the CD45 bright and CD14 negative population. The isotype control served as reference for negative events. Spectral compensation was established daily. Quality control included optimization for lymphocyte recovery, purity of analysis gate, lymphosum, and replicate determined according to CDC guidelines. In addition, measurement of intracellular cytotoxic protein concentrations was performed using quantitative fluorescence. Levels of intracellular perforin, granzyme A or granzyme B, conjugated to phycoerythrin within lymphocytes subsets, both NK cells and CD8+ T cells, were simultaneously assessed with a 5-color system using a maximum-yield protocol (Maher et al., 2002; Broderick et al., 2011). Summary statistics for these markers are listed at each phase of exercise for each subject group in Table S3.

2.1.5. Cytokine profiles

Plasma was also separated from each sample within 2 h of collection and stored at −80 °C until assayed. We measured 16 cytokines in plasma using Quansys reagents and instrument (Quansys

Biosciences, Logan, Utah). The Quansys Imager, driven by an 8.4 megapixel Canon 20D digital SLR camera, supports 96 well plate based chemiluminescent imaging. The Q-Plex™ Human Cytokine-Screen (16-plex) is a quantitative enzyme-linked immunoabsorbent assay (ELISA)-based test where sixteen distinct capture antibodies have been absorbed to each well of a 96-well plate in a defined array. The range of the standard curves and exposure time were adjusted previously to provide reliable comparisons between subject groups in this illness population at both low and high cytokine concentrations in plasma. Second order polynomial regression models were used as standard calibration curves. Quadruplicate determinations were made, i.e., each sample was run in duplicate in two separate assays. Statistics reported previously in Broderick et al. (2010) indicated an average coefficient of variability (CV) of 0.20 for inter-assay and 0.09 for intra-assay repeatability. Summary statistics for cytokine concentrations at each phase for each subject group are listed in Table S3.

2.2. Numerical analysis

2.2.1. Comparative analysis of gene expression

Prior to analysis raw microarray data was corrected for chip-to-chip variability. Removal of background signal intensity was performed using GeneChip RMA (GC-RMA), an extension of classical Robust Multichip Average (RMA) algorithm (Wu et al., 2004; Wu and Irizarry, 2005; Katz et al., 2006). The range of intensity values were adjusted using quantile normalization (Bolstad et al., 2003). Expression values were transformed logarithmically (log2) to improve normality. The significance of changes in gene expression across subject groups and across all 3 time-points was evaluated using a 2-way analysis of variance (ANOVA-2). In all cases, null probability values were adjusted for multiple comparisons using the method proposed by Storey (2002) for estimating the false discovery rate (FDR) and probability of false discovery q. Gene and pathway associations of Affymetrix probe sets were obtained using the Protein Analysis THrough Evolutionary Relationships (PANTHER) software system (Mi et al., 2010).

2.2.2. Estimating pathway activity

Individual gene products are expressed in a highly coordinated manner that supports the function of biochemical reaction pathways in the cell. In accordance with this we have used a novel algorithm developed by Efroni et al. (2007, 2008) to estimate the activity of known pathway segments from the expression of the genes that encode their components. The first step consists of converting a continuous measure of gene expression into a discrete gene status, namely up-expressed or down-expressed. In brief, for every individual gene the distribution of expression values across all samples are numerically fit to two separate gamma probability distribution functions: one describing the distribution of expression values supporting the up state and one describing the distribution of values supporting the down state. The expectation maximization (EM) algorithm is applied to provide a set of maximum-likelihood estimates for the parameters a_i and b_i as well as the values of the mixture coefficients η_i that weigh the contribution of the gamma distribution (Eqs. 1–3) for each state $i = 1,2$. The vector of these parameters is estimated iteratively from Θ^0 to Θ such that the function Q (Θ, Θ^0) is maximized (Eq. 1).

$$Q(\theta, \theta^0) = \sum_t \sum_i \omega_{t,i}(\log(\eta_i) - \log(\gamma(x_t; a_i, b_i)) \tag{1}$$

where

$$\gamma(x_t; a_i, b_i) = \frac{1}{b_i^{a_i}\Gamma(a_i)} x_t^{a_i - 1} e^{\frac{x_t}{b_i}} \tag{2}$$

$$\omega_{t,i} = \frac{\eta_i^0 \gamma(x_t; a_i^0, b_i^0)}{\sum_j \eta_j^0 \gamma(x_t; a_i^0, b_i^0)} \tag{3}$$

The probability of a gene being up-expressed given its expression level x is the probability of occurrence of the "up" state overall $p_{UP} = N_{UP}/N$ multiplied by the probability of expression level x corresponding to an up-expressed state or $\gamma(x; a_{UP}, b_{UP})$; where N_{UP} is the number of genes that are up-expressed among all N genes. It is important to note that this manipulation does not assign a discrete up or down state to a gene but instead provides a continuous scale of expression that includes being neither up nor down-expressed.

Every pathway consists of a collection of reaction steps or interactions. These can be modeled as logic functions with genes serving as inputs and outputs. In the current protocol the activation level of such a logical function was computed based on the joint conditional probability that the input genes $k \in$ set I are in an up-expressed state based on measured gene expression (Eq. 4). As we have estimated the corresponding probability that the set of output genes $k \in O$ are also in an up-expressed state (Eq. 5), we can use the agreement between input and output as a measure of consistency C_s for the activation of reaction step s (Eq. 6).

$$p(s = \text{active}) = \prod_{k \in 1} p(g_k = \text{“up”}) \tag{4}$$

$$p(O = \text{“up”}) = \prod_{k \in O} p(g_k = \text{“up”}) \tag{5}$$

$$C_s = p(s = active) \times p(O = \text{“up”}) + (1 - p(s = active)) \times (1 - p(O = \text{“up”})) \tag{6}$$

Using the approach outlined in Eqs. (1)–(6), activation levels and consistency were calculated in each patient sample for all reaction steps described in 582 candidate pathways aggregated from the National Cancer Institute (NCI)–Nature Pathway Interaction Database (PID) (Schaefer et al., 2008) and the Kyoto Encyclopedia of Genes and Genomes (KEGG) database (Kanehisa et al., 2010). The NCI-Nature PID database is itself an aggregation of 135 pathways curated by the NCI-Nature team with an additional 322 pathways imported from the BioCarta (www.biocarta.com) and Reactome (Croft et al., 2011; Matthews et al., 2009) databases. In each individual sample the activity of each pathway was calculated as the average activation level across all component interactions. Activation scores in each subject group were log transformed to improve normality and compared for each pathway at each time point using both parametric (t test) and non-parametric (Wilcoxon rank sum) tests. Once again, two-way analysis of variance (ANOVA-2) was used to assess the significance of group, time and group × time interactions with the false discovery rate (FDR) estimated using Storey (2002). The potential of each pathway activation score as a diagnostic marker was described in terms of receiver-operating characteristics (ROC) by the increase above random in the area under the curve (AUC > 0.50). All the computations were conducted with the MATLAB software environment (The MathWorks Inc., Natick, MA).

2.2.3. Identifying patterns of pathway-symptom interaction

To describe in statistical terms the correlation of pathway activity with symptom severity via changes in immune cell population, function and signaling we constructed multi-layer empirical networks using partial linear correlation as a measure of association (Emmert-Streib, 2007; Magwene and Kim, 2004). This measure adjusts the pair-wise correlation of marker x_i with marker x_j for the indirect correlation contributed by a confounding marker x_k.

$$r(x_i, x_j|x_k) = \frac{r(x_i, x_j) - r(x_i, x_k)r(x_j, x_k)}{\sqrt{(1 - r^2(x_i, x_k))}\sqrt{(1 - r^2(x_j, x_k))}} \quad (7)$$

In order to cast each pair-wise association in the proper context these were adjusted for age and BMI within the GWI subject group. We summarized the contribution of these confounding effects by computing the first principal component for these two markers and using this as the context variable x_k (Eq. 7). Null probability p was computed by transforming the correlation to create a t statistic having $n-2$ degrees of freedom for n observations. Confidence bounds were based on an asymptotic normal distribution of $0.5*\log((1 + r(x_i x_j\,|x_k))/(1-r(x_i x_j\,|x_k)))$, with an approximate variance equal to $1/(n-3)$ when variables have a multivariate normal distribution. Once again the probability of false discovery q was estimated according to Storey (2002) with $q < 0.05$ indicating significant associations.

Based on this measure of association, multi-level hierarchical networks were constructed in GWI linking pathway activation to symptom severity through changes in immune cell abundance, function and signaling. Minimum paths $p_{i\ j}$ linking each pathway x_i to each symptom y_j were computed and subjected to a logical consistency constraint (Eq. 8). This stipulated that the direction of change in the symptom score in GWI, sign(Δy_j), must be consistent with the direction of change in pathway activation, sign(Δx_i), when propagated along the sequence of correlation coefficients $r_{n\ m}$ associated with each segment [n,m] of the shortest path $p_{i\ j}$.

$$sign(\Delta x_i) = sign(\Delta y_j) \prod_{n.m \in p} sign(r_{nm}) \quad (8)$$

All calculations related to network identification and rationalization, including minimum path calculations, were conducted with the MATLAB software environment (The MathWorks Inc., Natick, MA). The graphical rendering of correlation networks as well as the analysis of network attributes was done using the Cytoscape platform (Smoot et al., 2011) with the NetworkAnalyzer component (Assenov et al., 2008).

3. Results

3.1. Differential expression of transcription-associated markers

The number of microarray probe sets changing in expression across time, subject group, or showing a group × time interaction with a probability of false discovery $q < 0.05$ are shown in Table 2. Over 1000 probes sets (1080) differed significantly in expression across time in GWI. Using the PANTHER database this corresponded to 644 genes associated with 79 pathways. For the CFS/ME disease control group we found close to 300 probe sets (280) changing across time. These mapped to 180 genes that were in turn associated with 30 pathways. These changes across time were dwarfed by the very broad differences in expression observed across subject groups. Close to 9000 probe sets (8819) were differentially expressed in GWI across all time points compared to the healthy control group. These mapped to 4404 genes associated with 126 pathways in the PANTHER database. In CFS/ME over 10,000 probe sets (10,222) changed significantly in expression compared with the healthy control group. These probe sets corresponded to 4963 genes that in turn mapped onto 132 pathways. While significant effects were found for both time and group no probe sets presented significant time × group interactions with $q < 0.05$.

3.2. Preferential activation of known pathways

Using the methodology proposed by Efroni et al. (2007, 2008) we obtained estimates of pathway activation levels for each sample. As with the previous probe-level analysis, a 2-way ANOVA of pathway activation found no significant time–group interactions. However we also found that by enforcing pathway regulatory logic and aggregating across multiple step reactions, changes across time were no longer discernable. As shown in Table 2, only group effects were significant at $q < 0.05$. In GWI, 127 pathways differed significantly in activation level compared to the healthy control group. This is comparable with the number found by relying on the gene and pathway annotation of the probe sets. The changes in the predicted activation level of 19 of these pathways were found only in GWI and exceeded by 10% the activation level found in the healthy control group for that same pathway (Table 3). Results indicated decreased activation in several key elements of apoptotic signaling. For example, we found reduced pathway activity for tumor necrosis factor receptor 1 (TNFr1) and FAS/CD95 receptor signaling as well as tyrosine kinase mediated activation of T cell receptor signaling and thrombospondin-1 (TSP-1) mediated induction of cell death. This also included decreased kinase A-anchoring protein AKAP95 activity and its involvement via caspase-3 in the apoptotic caspase cascade. This was accompanied by reduced activity in several transcriptional regulatory pathways. In particular we found reduced activation of nuclear receptors heterodimers retinoic acid receptor (RAR) and retinoid X receptor (RXR). In the absence of ligand, DNA-bound RXR/RARA are known to represses transcription via co-repressors NCOR1, SMRT (NCOR2) and histone deacetylase (HDAC). Also disrupted was activation of RAs-related Nuclear protein (RAN), an androgen receptor (AR) co-activator, via the RAN binding protein, as well as activation of RNA polymerase III, a decisional regulator of cell survival during stress.

In contrast, we observed increased activation of pathways involving broad-acting immune modulator NF-κB (nuclear factor kappa-light-chain-enhancer of activated B cells) (Figure S1). We also found significantly elevated visual signal transduction and activation of a KEGG super-pathway describing broad-scale receptor–ligand interactions in neurotransmission. This was accompanied by increased biosynthesis of phenylpropanoid compounds, a process promoted by the dopamine precursor phenylalanine. Finally, with possible ties to environmental exposure, we also observe chronic activation of the detoxification pathway supporting degradation of the androgen-disrupting hydrocarbon bisphenol.

While CFS/ME and GWI are commonly considered sister illnesses, we found little overlap at the level of pathway activation. Specifically, we found 25 pathways that were uniquely expressed in CFS/ME ($q < 0.05$) and exceeded by 10% the corresponding activation level found in the healthy control group (Table S4). These typically involved suppression of transcriptional regulation and cell cycle. Changes in energy metabolism included activation of antioxidant lipoic acid metabolism accompanied by decreases in caffeine-induced lipolysis and leptin-mediated control of insulin resistance. These changes coincided with decreased B cell receptor signaling. Only 8 pathways were shared by both CFS/ME and GWI ($q < 0.05$) with activation ranging from 93% to 110% of healthy control (Table S5). Control of skeletal myogenesis calmodulin-dependent kinase (CaMK) and granzyme A mediated apoptosis were the most up and down-expressed respectively.

In a first partial validation, data collected beforehand for quality control were examined for concordance with these results. A survey of 10 genes selected a priori for their involvement in immune signaling was conducted using the NanoString nCounter system. Results of a 2-way ANOVA indicated significant group effects for the expression of beta actin (ACTB), fibroblast growth factor-binding protein 2 (FGFBP2), acidic ribosomal protein P0 (RPLP0) as well as zinc finger and BTB domain-containing protein 16 (ZBTB16). This was especially noticeable in GWI subjects at peak effort (T1) for RPLP0 and ZBTB16 expression. Pathway annotations listed in

Table 2
Summary of microarray probe sets with significant difference in expression level in GWS and CFS/ME groups compared to the healthy control group. Significance was based on *p*-values for time, group and time × group effects estimated from a 2-way ANOVA and set as $p < 0.05$. Adjustment for multiple comparisons was based on Storey's *q*-statistic ($q < 0.05$).

	p Time < 0.05	*q* < 0.05	*p* Group < 0.05	*q* < 0.05	p Time × group < 0.05	*q* < 0.05
Number of probe sets						
GWI vs HC all times	5313	1080	14,510	8819	583	0
CFS vs HC all times	5008	280	15,185	10,222	3364	0
Number of genes (Unigene IDs mapped)						
GWI vs HC all times	2894	644	6682	4404	320	0
CFS vs HC all times	2951	180	6914	4963	1972	0
Number of pathways (from mapped IDs)	Panther Database					
GWI vs HC all times	114	79	138	126	65	0
CFS vs HC all times	112	30	137	132	109	0
Number of pathways (PID path activation)	NCI/Nature PID					
GWI vs HC all times	15	0	135	127	5	0
CFS vs HC all times	45	0	128	89	39	0

Table 3
Significant differences (ANOVA-2) in inferred pathway activity unique to GWI ($q < 0.05$) with a mean absolute fold change (FC) in pathway activity in excess of 10% compared to healthy control subjects (HC) (1.10 < pathway activity FC < 0.90). Performance as potential diagnostic markers is shown as area under receiver–operator characteristic (ROC) curve (AUC) in excess of 0.50, the value expected from a random assignment.

Node #	Pathway #	Pathway name	*p* Group	*q* Value	ROC AUC > 0.50	Pathway activity FC (GWI:HC)
74	10	AKAP95 role in mitosis and chromosome dynamics (BioCarta)	0.000	0.006	0.173	0.75
497	433	TSP-1 induced apoptosis in microvascular endothelial cell (BioCarta)	0.001	0.007	0.127	0.86
419	355	RNA polymerase III transcription (BioCarta)	0.001	0.010	0.090	0.86
293	229	LCK and FYN tyrosine kinases in initiation of TCR activation (BioCarta)	0.003	0.021	0.072	0.86
441	377	Signaling events mediated by HDAC class III (NCI/Nature)	0.009	0.030	0.083	0.88
300	236	Lissencephaly gene (LIS1) in neuronal migration and development (NCI/Nature)	0.043	0.038	0.101	0.89
196	132	FAS signaling pathway (CD95) (BioCarta)	0.004	0.016	0.102	0.89
138	74	Caspase cascade in apoptosis (BioCarta)	0.000	0.004	0.159	0.89
486	422	TNFr1 signaling pathway (BioCarta)	0.003	0.011	0.083	0.89
197	133	FAS signaling pathway (CD95)(NCI/Nature)	0.001	0.007	0.157	0.89
307	243	MAP kinase inactivation of SMRT corepressor (BioCarta)	0.024	0.042	0.114	0.90
431	367	RXR and RAR heterodimerization with other nuclear receptor (NCI/Nature)	0.009	0.038	0.071	0.90
120	56	Bisphenol a degradation (KEGG)	0.007	0.022	0.128	1.11
329	265	Neuroactive ligand–receptor interaction (KEGG)	0.004	0.014	0.133	1.12
458	394	Stress induction of HSP regulation (BioCarta)	0.000	0.006	0.195	1.13
369	305	Phenylpropanoid biosynthesis (KEGG)	0.016	0.032	0.136	1.18
102	38	Atypical NF-κB pathway (NCI/Nature)	0.001	0.006	0.172	1.21
513	449	Visual signal transduction: cones (NCI/Nature)	0.032	0.044	0.111	1.26
333	269	NF-κB signaling pathway (BioCarta)	0.000	0.006	0.170	1.33

Table S2B for ACTB and ZBTB16 show an overlap with the results in Tables 3 and S5, in particular with regard to ACTB's role in nAChR signaling and its subsequent up-regulation of NF-κB activity (Chernyavsky et al., 2010) via CaMK. Likewise changes in ZBTB16 aligned with predicted changes in co-repressor SMRT activity and its role in facilitating the recruitment of histone deacetylases (HDAC).

3.3. Pathway associations with symptom severity in GWI

The pathways described above are not expressed independently of one another but instead belong to a well-integrated biochemical network that both drives and responds to changes across multiple levels of biology. To explore how these changes in pathway activation might be relevant to GWI we examined their correlation with changes in 24 indicators of symptom severity measured at rest, corrected for age and body–mass index (BMI) (Table S1). Rather than examine direct association between pathway and symptom, we introduced intermediate layers describing changes 12 lymphocyte sub-populations, 7 measures of cell cytotoxicity, and the expression of 16 cytokines, NPY and cortisol. A change in pathway activation was considered relevant to illness severity if it first correlated significantly changes in immune cell demographics and function, then cell–cell signaling. We also imposed a consistency constraint whereby the direction of correlation along this path had to align with differences in pathway and symptom expression observed across subject groups. This analysis of cell and protein profiling was used to provide additional validation, limit spurious associations and improve biological interpretation.

Using these constraints multi-level networks were constructed from 518 candidate intracellular, cellular, intercellular and symptom-level nodes. As some features of GWI might correlate best with exercise capacity we constructed separate networks linking baseline symptom severity with these markers at rest (T0), peak effort (T1) and at recovery (4 h after peak effort) (T2). Only associations with a partial correlation coefficient $|r| > 0.10$ and a significance of $q < 0.05$ were conserved. From these networks we extracted the shortest paths linking each symptom measure to the 19 pathways significantly expressed in GWI (Table 3). When enforcing hierarchical and significance constraints, no sub-networks were found linking symptom expression to GWI pathways at rest. However at peak effort and in the recovery phase all 19 pathways were linked to symptom severity by between 4 and 12 edges (Tables S6 and 7), with correlation strength typically exceeding $|r| = 0.50$ (Figure S2).

At peak effort this network consisted of 23 symptom nodes separated from the 19 GWI pathway nodes by a layer of 9 soluble

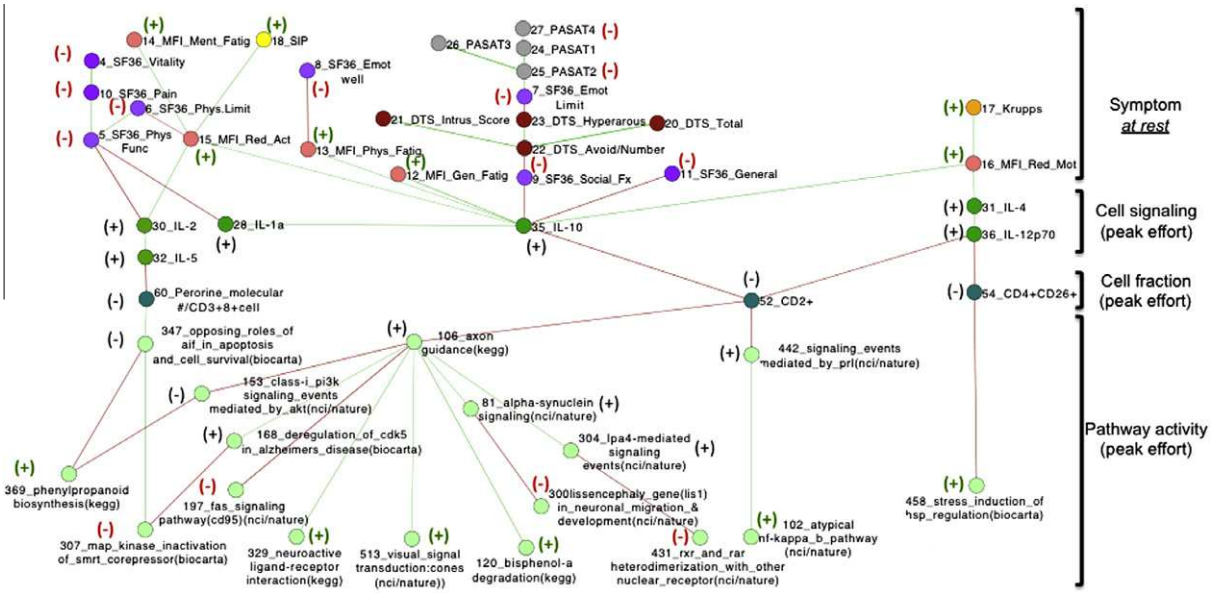

Fig. 1. GWI pathway sub-network interface with symptom severity at peak effort. Association sub-network in GWI at peak effort (T1) during exercise showing only uninterrupted paths linking symptom nodes with pathway nodes, through layers containing cytokine/hormone and cell abundance/toxicity nodes. Links or edges indicate pair-wise partial correlations with a probability of false discovery $q \leqslant 0.05$; positive correlations are shown as green edges, negative correlations are shown as red edges. Bold green (+) and bold red (−) indicate measured statistically significant increases or decreases in expression respectively, while the same symbols in black font indicate estimated direction of change based on the consistency constraint of Eq. (8). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

mediators (cytokines, NPY and cortisol), 6 cell population–function nodes and 10 intermediate pathway nodes. A similar network was produced at recovery. Enforcing logical consistency propagated downward from symptom severity resulted in rationalization of this network leaving 10 of the original 19 GWI pathways (Fig. 1) at peak effort only. No pathways satisfied this constraint at recovery (T2). Scores for DTS, PASAT as well as SF36 social function and emotional limit clustered together as a sub-network linked to IL-10 levels. Interestingly though IL-10 was significantly lower in GWI as a group, within-group increases aligned with increased illness severity (Figure S3). Similarly SF36 measures for physical function, physical limit, pain, and vitality cluster together and were linked to IL-1α, IL-2 and IL-5 levels. MFI indicators were more diffuse in their association with reduced motivation, for example, standing apart and correlating positively with changes in IL-4, 12 and IL-10. SF36 general score also stood apart and correlated negatively with levels of IL-10. Of the 6 cytokines that remained in the sub-network, IL-10 had the broadest effects on severity with direct connections to 6 symptom constructs including SF36 social function which itself supported a cluster of 10 symptoms.

Within-group changes in IL-10 and IL-1α levels, along with IL-4 and IL-12, aligned with a shift in the relative abundance of CD2+ T and NK lymphocytes. Similarly changes in IL-2 and IL-5 aligned with changes in intracellular perforin in cytotoxic T cells (CD3+CD8+). Both branches correlated with expression of the axon guidance pathway (Fig. 1), a hub linking 8 of the original 19 GWI pathways under the broad theme of neuronal development and migration. Aligning with results from the nString panel, two of these 8 satellite pathways involved regulation of transcriptional co-repressor SMRT, and its associated activation for RAR and RXR heterodimers. The latter were linked to axon guidance via the regulation of Cdk5 and LPA4 signaling respectively, two pathways associated with neuronal maturation and migration. Also consistent with this theme were pathway nodes for regulation of lissencephaly gene LIS-1 and α-synuclein signaling as well as neuroactive ligand–receptor signaling and visual signal transduction, the second most over-expressed GWI pathway. This cluster centered about the axon guidance pathway was tethered to cytotoxic T cell perforine levels via apoptosis-inducing factor (AIF) activity, which in turn was linked to SMRT co-repressor activity and biosynthesis of phenypropanoid, an inhibitor of apoptosis. The latter was also linked to modulation of PI3K/AKT/mTOR activity. In parallel with this major hub surrounding axon guidance, a second minor branch from the CD2+ cell population node consisted of 2 pathways associated with androgen-mediated regulation of NF-κB activation. Finally, changes in the relative abundance of Th17 cells (CD4+CD26+) aligned with increased induction of heart shock protein (hsp) and the expression of IL-12, an NK cell primer.

4. Discussion

In this study we use a standard exercise challenge to stimulate immune and endocrine signaling in a group of veterans with Gulf War Illness (GWI), a disease control group of non-veterans with chronic fatigue syndrome (CFS/ME) and healthy control subjects. Circulating lymphocytes collected at initial rest, peak effort (VO2 max) and 4 h post-effort were analyzed for gene expression using both a conventional single-gene approach and a more novel approach whereby transcript levels were projected onto the regulatory circuitry of known pathways. This projection resulted in a loss of resolution with respect to time however strong group effects were found with 19 pathways being uniquely expressed in GWI compared to both healthy controls and CFS/ME, an illness with similar clinical presentation. Activity was suppressed in roughly two thirds of these pathways including TNF receptor (or death receptor) signaling and elements of apoptotic signal transduction. Although signatures obtained in CFS and GWI shared some commonalities, CFS presented with an almost opposite suppression of cell cycle, actin remodeling and metabolism. Calcium-mediated T cell activation, IFN signaling and B cell receptor activation were also suppressed in CFS. Indeed, calcium induced activation of calcineurin and subsequent calmodulin binding was a recurring theme among pathways expressed in CFS. Among the

latter we found control of skeletal myogenesis by CaMK, a process involving activation of P2X5 receptor (Ryten et al., 2002). Calmodulin and calcineurin-mediated signaling has also been implicated in desensitization of pain receptor TRPV1 (Numazaki et al., 2003). Involvement of purinergic P2X5, TRPV1 as well as IL-10 immune signaling has been substantiated in recent work by Light et al. (2012). However, as this group was very small ($n = 7$) a more adequate interpretation would be that the CFS subjects studied here could not be easily reconciled with GWI in terms of pathway expression.

General suppression of apoptotic signaling in GWI was accompanied by a significant increase in broad-scale ligand–receptor interactions supporting neurotransmission as well as increased activation of master transcriptional regulator NF-κB. Indeed, when these pathways were cast in the context of symptom severity and changes in immune cell population, function and signaling, the pathway mediating axon guidance arose as a major hub aggregating these effects with a number of signal transduction pathways including visual signal transduction and neuronal migration. Interestingly retinal axon branching and similar processes are promoted by neurotrophic factor (BDNF) and its receptor tropomyosin-related kinase B (Marler et al., 2008). In recent work Barbier et al. (2009) found alterations in gene expression for both BDNF and TrkB in stressed animals exposed to PB. Other pathways include LPA4 mediation of PI3K/AKT pathway (Lee et al., 2008; Yanagida et al., 2007) and signaling via α-synuclein, a key protein in Parkinson's disease (Tani et al., 2010). All support either directly or indirectly processes related neuronal migration. Although these measures of pathway activation were expressed in peripheral blood, similar profiles have been found in PBMC gene expression of other neuro-immune and neurodegenerative disorders. For example, a recent analysis of PBMC gene expression in subjects with Parkinson's disease reported similar disruption of pathways associated with axonal guidance, calcium signaling, death signaling and apoptosis (Mutez et al., 2011). Also found were altered activation of MAP kinase pathway and actin-mediated cytoskeleton formation. These results are consistent with our observations of altered FAS signaling, MAP kinase inactivation of SMRT and their link with the axon guidance hub pathway. Axon guidance was also one of the most significant pathways found in a functional enrichment analysis of the top 100 disease-related hub genes expressed in the PBMC of amyotrophic lateral sclerosis (ALS) patients (Saris et al., 2009). Differences in FAS-mediated apoptotic signaling were also found relevant to ALS pathogenesis (Raoul et al., 2006). Of note, results obtained in blood by Mutez et al. (2011) showed substantial alignment with gene expression in biopsies collected from the substantia nigra pars compacta of PD subjects, supporting the relevance of PBMC profiling in these studies. Similar evidence has been reported in the study of other neurological conditions including Huntington's and Alzheimer's disease (Runne et al., 2007; Maes et al., 2007).

In parallel with neuronal migration pathways we found activation of master transcriptional regulator nuclear factor-κB (NF-κB) in relation to prolactin (PRL) signaling. Up-regulation of the NF-κB/Rel family has been observed in monocytes of men (O'Donovan et al., 2011) and PBMC of women with PTSD (Pace et al., 2012). In animal models, selective inhibition of this transcription factor significantly reduced the prevalence of extreme PTSD-like response to stress (Cohen et al., 2011). Prolactin, a stress-responsive pituitary hormone, has been shown to increase the binding activity of NF-κB and interferon regulatory factor-1 (IRF-1), promoting TNF-α and IL-12 secretion (Brand et al., 2004). Secretion of IL-10 was also induced but only at high levels of PRL. A T-cell cytokine, prolactin is important in maintaining immune system homeostasis under stress and preventing glucocorticoid-induced lymphocyte cell death (Yu-Lee, 2002; Buckley, 2001). This role would be consistent with the down-regulation of death signaling found in our current analysis as well as recent evidence pointing to HPA axis dysfunction in these subjects, potentially involving altered pituitary function (Golier et al., 2007, 2009). Finally the role of PRL in mediating neurogenesis (Walker et al., 2012), astrocyte proliferation and associated IL-1α production (DeVito et al., 1995) may explain at least in part the elevated PRL levels found in autoimmune illnesses like systemic lupus erythematosus (SLE) and multiple sclerosis (MS) (Shelly et al., 2012). More importantly, this role might offer a tentative mechanistic basis supporting the association of GWI severity with PRL/NF-κB and neurogenic pathway activation, via IL-10 and IL-1α in particular.

Also consistent with known actions of prolactin, we found PRL-mediated NF-κB activity to be associated with GWI severity through changes in T and NK cell abundance (CD2+) as well as through IL-10 and IL-12 expression. As mentioned previously, IL-10 emerged as a hub cytokine that supported changes in virtually all GWI symptom measures along with IL-1α. Elevated levels of IL-10 have been associated with increased severity of depressive symptoms in MS (Heesen et al., 2005). Indeed, a recent gene set analysis performed on tissue samples from the prefrontal cortex of patients with major depressive disorder (MDD) suggested up-regulation of a variety of pro- and anti-inflammatory cytokines including those identified here as correlates of GWI severity (Fig. 1), namely IL-1α, IL-10, and IL-12 as well as IL-2 and IL-5 (Shelton et al., 2011; Schmidt et al., 2003). Not surprisingly, NF-κB signaling plays a key role in mediating the actions of IL-1 and stress in the onset of depression-like behavior (Koo et al., 2010). Changes in IL-1R signal transduction, one of only 2 pathways representing IL-1 signaling in this dataset, were statistically significant ($q = 0.045$) but the magnitude of changes in pathway activation were less than 10% of levels found in healthy control. In a previous analysis we found a strong influence of IL-1α on neuroendocrine-immune signaling in GWI even though levels in plasma did not differ significantly from control (Broderick et al., 2010). This same work also pointed to a muted cortisol response in GWI, which is consistent with over-expression of NF-κB activity (Wolf et al., 2009).

Though several of the markers implicated here are also responsive to exercise, e.g. T and NK cell abundance (Walsh et al., 2011), IL-1α (Kimura et al., 2001), IL10, prolactin release (Rojas Vega et al., 2012) as well as neurogenic (Walker et al., 2012) and NF-κB activation (Kim et al., 2009), it is important to remember that they were selected on the basis of changes in symptom severity and pathway activation across subject groups. As a result we found some aspects of normal exercise response to be over-expressed or expressed earlier than normal in GWI. For example, in GWI we found an excessive activation of normal acute IL-1α response (Kimura et al., 2001) with involvement of IL-10 (Walsh et al., 2011) and IL-4 (Rosa Neto et al., 2011) at peak effort when these are typically expressed much later. In addition, there were notable deviations from normal exercise response. Activity of cyclin-dependent kinase 5 (Cdk5) (Ghiani et al., 2007) is typically up-regulated during exercise while α-synuclein activity is down-regulated (Dimatelis et al., 2012). While we recovered the expected trend in Cdk5 activation, we predicted an opposite trend towards up-regulation of synuclein signaling in GWI. Similarly, we inferred an inverse trend in PI3K/AKT pathway activity, which is normally increased during exercise (Walsh et al., 2011). Though our analysis suggested decreasing CD2+ relative abundance consistent with exercise-induced lymphocytosis, we observed a concurrent decrease in several facets of apoptotic signaling as well as positive correlation of IL-2 with increasing severity (Gleeson and Bishop, 2005).

This work points to chronic NF-κB activation as a potentially key component of GWI. Though recent work in animal models sup-

ports the involvement of altered cholinergic-immune signaling in GWI as a result of exposure to acetylcholine (ACh) mediators (PB, organophosphates, sarin, etc.…), we did not observe direct evidence of this here. However it must be emphasized that our analysis was constrained to a specific set of documented pathways and was by no means comprehensive. For example, ACh signaling is known to be a significant modulator of immune function (Kawashima and Fujii, 2003) and activation of nicotinic acetylcholine receptors (nAChR) by SLURP-1 is known to induce of NF-κB expression (Chernyavsky et al., 2010). Blocking nAChR function significantly increased basal ACTH and corticosterone in animals and completely inhibited PRL response to stress (Weidenfeld et al., 1983). This is consistent with bi-directional ACh signaling in lymphocytes (Kawashima et al., 2012).

In addition to the limited scope of the pathways examined, we found the method of Efroni et al. (2007, 2008) to be conservative since significant up-expression or down-expression of a pathway requires a corresponding change in the majority of individual genes supporting that pathway. As a result, it almost certainly leads to the loss of more subtle signaling mechanisms. Indeed in this instance, there was a loss of temporal trends across the phases of exercise. This being said, we might also expect that those pathways identified in this analysis were well supported at the level of gene expression. The method also provides a much-needed framework for the functional interpretation of gene expression results. Many findings from this work would not have emerged from a conventional comparative analysis, emphasizing the importance of casting marker expression in the confirmatory context of multiple layers of biology. This is especially relevant in the study of complex disorders like GWI. Although the classification performance of pathway activation levels (AUC < 0.70) did not encourage their use single biomarkers, this was still sufficiently elevated to motivate future investigation of their use in combination as a biomarker panel for diagnosis and possible sub-typing of GWI (Haley et al., 1997).

In summary, we found statistical association of baseline GWI symptom burden with increased activation at peak effort in pathways supporting neuronal development along with down-regulation of apoptotic signaling. This was accompanied by prolactin-mediated increases in NF-κB activation, a shift in T and NK cell population and the expression of IL-10 and IL-1α principally. These changes not only exceeded normal exercise response but also differed in nature. NF-κB proteins have been detected in many chronic inflammatory and autoimmune conditions including cancer (Ben-Neriah and Karin, 2011). Though the link to ACh signal disruption was not substantiated directly, disruption of prolactin-mediated NF-κB activation in concert with altered signaling supporting neurogenesis and apoptosis merit further scrutiny. As this was an exploratory analysis, confirmatory profiling of gene expression along candidate pathways identified here, NF-κB in particular, should be undertaken using a high-resolution method such as PCR. In addition, pharmacological blockage of NF-κB activity is possible using anti-TNF, anti-IL-1 as well as IκB kinase (IKK) inhibitors (Ben-Neriah and Karin, 2011). These might constitute important avenues for therapy in GWI as well as providing a treatment-based approach to verifying the illness processes proposed in this analysis.

Authors' contributions

Conceived and designed the experiments: M.A.F., N.G.K., G.B. Performed the experiments: L.N, M.A.F., N.G.K., Z.B. Analyzed the data: R.B.H., G.B., S.V., S.E. Contributed reagents/materials/analysis tools: S.E., L.N., M.A.F. Wrote the paper: G.B., R.B.H., S.E., N.G.K., M.A.F.

Acknowledgments

This analysis was funded by grants from the US Department of Defense CMDRP program (N. G. Klimas, M. A. Fletcher, G. Broderick), Merit Awards from the US Department of Veterans Affairs (N. G. Klimas, M. A. Fletcher) and from the CFIDS Association of America (G. Broderick, N. G. Klimas, M. A. Fletcher).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.bbi.2012.11.007.

References

Amourette, C., Lamproglou, I., Barbier, L., Fauquette, W., Zoppe, A., Viret, R., Diserbo, M., 2009. Gulf War illness: effects of repeated stress and pyridostigmine treatment on blood-brain barrier permeability and cholinesterase activity in rat brain. Behav. Brain Res. 203 (2), 207–214.

Assenov, Y., Ramirez, F., Schelhorn, S.E., Lengauer, T., Albrecht, M., 2008. Computing topological parameters of biological networks. Bioinformatics 24 (2), 282–284.

Barbier, L., Diserbo, M., Lamproglou, I., Amourette, C., Peinnequin, A., Fauquette, W., 2009. Repeated stress in combination with pyridostigmine. Part II: changes in cerebral gene expression. Behav. Brain Res. 197 (2), 292–300.

Ben-Neriah, Y., Karin, M., 2011. Inflammation meets cancer, with NF-κB as the matchmaker. Nat. Immunol. 12 (8), 715–723.

Bergner, M., Bobbitt, R.A., Carter, W.B., Gilson, B.S., 1981. The Sickness Impact Profile: development and final revision of a health status measure. Med. Care 19 (8), 787–805.

Bolstad, B.M., Irizarry, R.A., Astrand, M., Speed, T.P., 2003. A comparison of normalization methods for high density oligonucleotide array data based on variance and bias. Bioinformatics 19 (2), 185–193.

Brand, J.M., Frohn, C., Cziupka, K., Brockmann, C., Kirchner, H., Luhm, J., 2004. Prolactin triggers pro-inflammatory immune responses in peripheral immune cells. Eur. Cytokine Netw. 15 (2), 99–104.

Broderick, G., Fuite, J., Kreitz, A., Vernon, S.D., Klimas, N., Fletcher, M.A., 2010. A formal analysis of cytokine networks in chronic fatigue syndrome. Brain Behav. Immun. 24 (7), 1209–1217.

Broderick, G., Kreitz, A., Fuite, J., Fletcher, M.A., Vernon, S.D., Klimas, N., 2011. A pilot study of immune network remodeling under challenge in Gulf War Illness. Brain Behav. Immun. 25 (2), 302–313.

Buckley, A.R., 2001. Prolactin, a lymphocyte growth and survival factor. Lupus 10, 684–690.

Buysse, D.J., Reynolds 3rd, C.F., Monk, T.H., Berman, S.R., Kupfer, D.J., 1989. The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research. Psychiatry Res. 28 (2), 193–213.

Byrnes, A., Jacks, A., Dahlman-Wright, K., Evengard, B., Wright, F.A., Pedersen, N.L., Sullivan, P.F., 2009. Gene expression in peripheral blood leukocytes in monozygotic twins discordant for chronic fatigue: no evidence of a biomarker. PLoS ONE 4 (6), e5805.

Chernyavsky, A.I., Arredondo, J., Galitovskiy, V., Qian, J., Grando, S.A., 2010. Upregulation of nuclear factor-kappaB expression by SLURP-1 is mediated by alpha7-nicotinic acetylcholine receptor and involves both ionic events and activation of protein kinases. Am. J. Physiol. Cell Physiol. 299 (5), C903–C911.

Chi, H., 2012. Regulation and function of mTOR signalling in T cell fate decisions. Nat. Rev. Immunol. 12 (5), 325–338, Review.

Cohen, H., Kozlovsky, N., Matar, M.A., Zohar, J., Kaplan, Z., 2011. The characteristic long-term upregulation of hippocampal NF-κB complex in PTSD-like behavioral stress response is normalized by high-dose corticosterone and pyrrolidine dithiocarbamate administered immediately after exposure. Neuropsychopharmacology 36 (11), 2286–2302.

Collins, J.F., Donta, S.T., Engel, C.C., Baseman, J.B., Dever, L.L., Taylor, T., Boardman, K.D., Martin, S.E., Wiseman, A.L., Feussner, J.R., 2002. The antibiotic treatment trial of Gulf War Veterans' Illnesses: issues, design, screening, and baseline characteristics. Control. Clin. Trials 23 (3), 333–353.

Croft, D., O'Kelly, G., Wu, G., Haw, R., Gillespie, M., Matthews, L., Caudy, M., Garapati, P., Gopinath, G., Jassal, B., Jupe, S., Kalatskaya, I., Mahajan, S., May, B., Ndegwa, N., Schmidt, E., Shamovsky, V., Yung, C., Birney, E., Hermjakob, H., D'Eustachio, P., Stein, L., 2011. Reactome: a database of reactions, pathways and biological processes. Nucleic Acids Res. 39 (Database issue), D691–D697.

Davidson, J.R., Book, S.W., Colket, J.T., Tupler, L.A., Roth, S., David, D., Hertzberg, M., Mellman, T., Beckham, J.C., Smith, R.D., Davison, R.M., Katz, R., Feldman, M.E., 1997. Assessment of a new self-rating scale for post-traumatic stress disorder. Psychol. Med. 27 (1), 153–160.

DeVito, W.J., Avakian, C., Stone, S., Okulicz, W.C., Tang, K.T., Shamgochian, M., 1995. Prolactin induced expression of interleukin-1 alpha, tumor necrosis factor-alpha, and transforming growth factor-alpha in cultured astrocytes. J. Cell Biochem. 57 (2), 290–298.

Dimatelis, J.J., Hendricks, S., Hsieh, J.H., Vlok, M.N., Bugarith, K., Daniels, W.M., Russell, V.A., 2012. Exercise partially reverses the effect of maternal separation

on hippocampal proteins in 6-hydroxydopamine lesioned rat brain. Exp. Physiol. [Epub ahead of print].
Efroni, S., Schaefer, C.F., Buetow, K.H., 2007. Identification of key processes underlying cancer phenotypes using biologic pathway analysis. PLoS ONE 2 (5), e425.
Efroni, S., Carmel, L., Schaefer, C.G., Buetow, K.H., 2008. Superposition of transcriptional behaviors determines gene state. PLoS ONE 3 (8), e2901.
Eisen, S.A., Kang, H.K., Murphy, F.M., Blanchard, M.S., Reda, D.J., Henderson, W.G., Toomey, R., Jackson, L.W., Alpern, R., Parks, B.J., Klimas, N., Hall, C., Pak, H.S., Hunter, J., Karlinsky, J., Battistone, M.J., Lyons, M.J.Gulf War Study Participating Investigators, 2005. Gulf War veterans' health: medical evaluation of a US cohort. Ann. Int. Med. 142, 881–890.
Emmert-Streib, F., 2007. The chronic fatigue syndrome: a comparative pathway analysis. J. Comput. Biol. 14 (7), 961–972.
Fukuda, K., Nisenbaum, R., Stewart, G., Thompson, W.W., Robin, L., Washko, R.M., Noah, D.L., Barrett, D.H., Randall, B., Herwaldt, B.L., Mawle, A.C., Reeves, W.C., 1998. Chronic multisymptom illness affecting Air Force veterans of the Gulf War. JAMA 280, 981–998.
Ghiani, C.A., Ying, Z., de Vellis, J., Gomez-Pinilla, F., 2007. Exercise decreases myelin-associated glycoprotein expression in the spinal cord and positively modulates neuronal growth. Glia 55 (9), 966–975.
Gleeson, M., Bishop, N.C., 2005. The T cell and NK cell immune response to exercise. Ann. Transplant. 10 (4), 43–48.
Golier, J.A., Schmeidler, J., Legge, J., Yehuda, R., 2007. Twenty-four hour plasma cortisol and adrenocorticotropic hormone in Gulf War Veterans: relationships to posttraumatic stress disorder and health symptoms. Biol. Psychiatry 62 (10), 1175–1178.
Golier, J.A., Schmeidler, J., Yehuda, R., 2009. Pituitary response to metyrapone in Gulf War veterans: relationship to deployment. PTSD and unexplained health symptoms. Psychoneuroendocrinology 34 (9), 1338–1345.
Golomb, B.A., 2008. Acetylcholinesterase inhibitors and Gulf War illnesses. Proc. Natl. Acad. Sci. USA 105 (11), 4295–4300.
Gronwall, D.M., 1977. Paced auditory serial-addition task: a measure of recovery from concussion. Percept. Mot. Skills 44 (2), 367–373.
Haley, R.W., 1997. Is Gulf War syndrome due to stress? The evidence reexamined. Am. J. Epidemiol. 146 (9), 695–703.
Haley, R.W., Kurt, T.L., Hom, J., 1997. Is there a Gulf War Syndrome? Searching for syndromes by factor analysis of symptoms. JAMA 277 (3), 215–222, Erratum in JAMA 1997, 278(5), 388.
Haley, R.W., Spence, J.S., Carmack, P.S., Gunst, R.F., Schucany, W.R., Petty, F., Devous Sr., M.D., Bonte, F.J., Trivedi, M.H., 2009. Abnormal brain response to cholinergic challenge in chronic encephalopathy from the 1991 Gulf War. Psychiatry Res. 171 (3), 207–220.
Heesen, C., Koehler, G., Gross, R., Tessmer, W., Schulz, K.H., Gold, S.M., 2005. Altered cytokine responses to cognitive stress in multiple sclerosis patients with fatigue. Mult. Scler. 11 (1), 51–57.
Hurwitz, B.E., Coryell, V.T., Parker, M., Martin, P., Laperriere, A., Klimas, N.G., Sfakianakis, G.N., Bilsker, M.S., 2010. Chronic fatigue syndrome: illness severity, sedentary lifestyle, blood volume and evidence of diminished cardiac function. Clin. Sci. (Lond). 118 (2), 125–135.
Jonsdottir, I.H., 2000. Special feature for the Olympics: effects of exercise on the immune system: neuropeptides and their interaction with exercise and immune function. Immunol. Cell Biol. 78 (5), 562–570, Review.
Kanehisa, M., Goto, S., Furumichi, M., Tanabe, M., Hirakawa, M., 2010. KEGG for representation and analysis of molecular networks involving diseases and drugs. Nucleic Acids Res. 38, D355–D360.
Kang, H.K., Natelson, B.H., Mahan, C.M., Lee, K.Y., Murphy, F.M., 2003. Post-traumatic stress disorder and chronic fatigue syndrome-like illness among Gulf War veterans: a population-based survey of 30,000 veterans. Am. J. Epidemiol. 157, 141–148.
Katz, S., Irizarry, R.A., Lin, X., Tripputi, M., Porter, M.W., 2006. A summarization approach for Affymetrix GeneChip data using a reference training set from a large, biologically diverse database. BMC Bioinformatics 7, 464.
Kawashima, K., Fujii, T., 2003. The lymphocytic cholinergic system and its contribution to the regulation of immune activity. Life Sci. 74 (6), 675–696, Review.
Kawashima, K., Fujii, T., Moriwaki, Y., Misawa, H., 2012. Critical roles of acetylcholine and the muscarinic and nicotinic acetylcholine receptors in the regulation of immune function. Life Sci. 91 (21–22), 1027–1032.
Kerr, J.R., Petty, R., Burke, B., Gough, J., Fear, D., Sinclair, L.I., Mattey, D.L., Richards, S.C., Montgomery, J., Baldwin, D.A., Kellam, P., Harrison, T.J., Griffin, G.E., Main, J., Enlander, D., Nutt, D.J., Holgate, S.T., 2008a. Gene expression subtypes in patients with chronic fatigue syndrome/myalgic encephalomyelitis. J. Infect. Dis. 197 (8), 1171–1184.
Kerr, J.R., Burke, B., Petty, R., Gough, J., Fear, D., Mattey, D.L., Axford, J.S., Dalgleish, A.G., Nutt, D.J., 2008b. Seven genomic subtypes of chronic fatigue syndrome/myalgic encephalomyelitis: a detailed analysis of gene networks and clinical phenotypes. J. Clin. Pathol. 61 (6), 730–739.
Kim, S.Y., Jun, T.W., Lee, Y.S., Na, H.K., Surh, Y.J., Song, W., 2009. Effects of exercise on cyclooxygenase-2 expression and nuclear factor-kappaB DNA binding in human peripheral blood mononuclear cells. Ann. NY Acad. Sci. 1171, 464–471.
Kimura, H., Suzui, M., Nagao, F., Matsumoto, K., 2001. Highly sensitive determination of plasma cytokines by time-resolved fluoroimmunoassay; effect of bicycle exercise on plasma level of interleukin-1 alpha (IL-1 alpha), tumor necrosis factor alpha (TNF alpha), and interferon gamma (IFN gamma). Anal. Sci. 17 (5), 593–597.
Koo, J.W., Russo, S.J., Ferguson, D., Nestler, E.J., Duman, R.S., 2010. Nuclear factor-kappaB is a critical mediator of stress-impaired neurogenesis and depressive behavior. Proc. Natl. Acad. Sci. USA 107 (6), 2669–2674.
Krupp, L.B., LaRocca, N.G., Muir-Nash, J., Steinberg, A.D., 1989. The fatigue severity scale. Application to patients with multiple sclerosis and systemic lupus erythematosus. Arch. Neurol. 46, 1121–1123.
Lee, Z., Cheng, C.T., Zhang, H., Subler, M.A., Wu, J., Mukherjee, A., Windle, J.J., Chen, C.K., Fang, X., 2008. Role of LPA4/p2y9/GPR23 in negative regulation of cell motility. Mol. Biol. Cell. 19 (12), 5435–5445.
Light, A.R., Bateman, L., Jo, D., Hughen, R.W., Vanhaitsma, T.A., White, A.T., Light, K.C., 2012. Gene expression alterations at baseline and following moderate exercise in patients with Chronic Fatigue Syndrome and Fibromyalgia Syndrome. J. Int. Med. 271 (1), 64–81.
Liu, P., Aslan, S., Li, X., Buhner, D.M., Spence, J.S., Briggs, R.W., Haley, R.W., Lu, H., 2011. Perfusion deficit to cholinergic challenge in veterans with Gulf War Illness. Neurotoxicology 32 (2), 242–246.
Mach, M., Grubbs, R.D., Price, W.A., Nagaoka, M., Dubovický, M., Lucot, J.B., 2008. Delayed behavioral and endocrine effects of sarin and stress exposure in mice. J. Appl. Toxicol. 28 (2), 132–139.
Maes, O.C., Xu, S., Yu, B., Chertkow, H.M., Wang, E., Schipper, H.M., 2007. Transcriptional profiling of Alzheimer blood mononuclear cells by microarray. Neurobiol. Aging 28 (12), 1795–1809.
Magwene, P.M., Kim, J., 2004. Estimating genomic coexpression networks using first-order conditional independence. Genome Biol. 5 (12), R100.
Maher, K.J., Klimas, N.G., Hurwitz, B., Schiff, R., Fletcher, M.A., 2002. Quantitative fluorescence measures for determination of intracellular perforin content. Clin. Diagn. Lab. Immunol. 9 (6), 1248–1252.
Marler, K.J., Becker-Barroso, E., Martínez, A., Llovera, M., Wentzel, C., Poopalasundaram, S., Hindges, R., Soriano, E., Comella, J., Drescher, U., 2008. A TrkB/EphrinA interaction controls retinal axon branching and synaptogenesis. J. Neurosci. 28 (48), 12700–12712.
Matthews, L., Gopinath, G., Gillespie, M., Caudy, M., Croft, D., de Bono, B., Garapati, P., Hemish, J., Hermjakob, H., Jassal, B., Kanapin, A., Lewis, S., Mahajan, S., May, B., Schmidt, E., Vastrik, I., Wu, G., Birney, E., Stein, L., D'Eustachio, P., 2009. Reactome knowledgebase of human biological pathways and processes. Nucleic Acids Res. 37, D619–D622.
McArdle, W.D., Katch, F.I., Katch, V.L., 2007. Exercise Physiology: Energy, Nutrition, and Human Performance. Lippincott Williams & Wilkins, London, p. 954.
Mi, H., Dong, Q., Muruganujan, A., Gaudet, P., Lewis, S., Thomas, P.D., 2010. PANTHER version 7: improved phylogenetic trees, orthologs and collaboration with the Gene Ontology Consortium. Nucleic Acids Res. 38 (Database issue), D204–D210.
Mutez, E., Larvor, L., Leprêtre, F., Mouroux, V., Hamalek, D., Kerckaert, J.P., Pérez-Tur, J., Waucquier, N., Vanbesien-Mailliot, C., Duflot, A., Devos, D., Defebvre, L., Kreisler, A., Frigard, B., Destée, A., Chartier-Harlin, M.C., 2011. Transcriptional profile of Parkinson blood mononuclear cells with LRRK2 mutation. Neurobiol. Aging 32 (10), 1839–1848.
Numazaki, M., Tominaga, T., Takeuchi, K., Murayama, N., Toyooka, H., Tominaga, M., 2003. Structural determinant of TRPV1 desensitization interacts with calmodulin. Proc. Natl. Acad. Sci. U.S.A. (United States) 100 (13), 8002–8006.
O'Donovan, A., Sun, B., Cole, S., Rempel, H., Lenoci, M., Pulliam, L., Neylan, T., 2011. Transcriptional control of monocyte gene expression in post-traumatic stress disorder. Dis. Markers 30 (2–3), 123–132.
Pace, T.W., Wingenfeld, K., Schmidt, I., Meinlschmidt, G., Hellhammer, D.H., Heim, C.M., 2012. Increased peripheral NF-κB pathway activity in women with childhood abuse-related posttraumatic stress disorder. Brain Behav Immun 26 (1), 13–17.
Raoul, C., Buhler, E., Sadeghi, C., Jacquier, A., Aebischer, P., Pettmann, B., Henderson, C.E., Haase, G., 2006. Chronic activation in presymptomatic amyotrophic lateral sclerosis (ALS) mice of a feedback loop involving Fas, Daxx, and FasL. Proc. Natl. Acad. Sci. USA 103 (15), 6007–6012.
Reeves, W.C., Lloyd, A., Vernon, S.D., Klimas, N., Jason, L.A., Bleijenberg, G., Evengard, B., White, P.D., Nisenbaum, R., Unger, E.R.International Chronic Fatigue Syndrome Study Group, 2003. Identification of ambiguities in the 1994 chronic fatigue syndrome research case definition and recommendations for resolution. BMC Health Serv. Res. 3 (1), 25.
Rojas Vega, S., Hollmann, W., Strüder, H.K., 2012. Influences of exercise and training on the circulating concentration of prolactin in humans. J. Neuroendocrinol. 24 (3), 395–402.
Rosa Neto, J.C., Lira, F.S., Zanchi, N.E., Oyama, L.M., Pimentel, G.D., Santos, R.V., Seelaender, M., Oller do Nascimento, C.M., 2011. Acute exhaustive exercise regulates IL-2, IL-4 and MyoD in skeletal muscle but not adipose tissue in rats. Lipids Health Dis. 10, 97.
Runne, H., Kuhn, A., Wild, E.J., Pratyaksha, W., Kristiansen, M., Isaacs, J.D., Régulier, E., Delorenzi, M., Tabrizi, S.J., Luthi-Carter, R., 2007. Analysis of potential transcriptomic biomarkers for Huntington's disease in peripheral blood. Proc. Natl. Acad. Sci. USA 104 (36), 14424–14429.
Ryten, M., Dunn, P.M., Neary, J.T., Burnstock, G., 2002. ATP regulates the differentiation of mammalian skeletal muscle by activation of a P2X5 receptor on satellite cells. J. Cell Biol. 158 (2), 345–355.
Saris, C.G., Horvath, S., van Vught, P.W., van Es, M.A., Blauw, H.M., Fuller, T.F., Langfelder, P., DeYoung, J., Wokke, J.H., Veldink, J.H., van den Berg, L.H., Ophoff, R.A., 2009. Weighted gene co-expression network analysis of the peripheral blood from Amyotrophic Lateral Sclerosis patients. BMC Genomics 10, 405.

Schaefer, C.F., Anthony, K., Krupa, S., Buchoff, J., Day, M., Hannay, T., Buetow, K.H., 2008. PID: the Pathway Interaction Database. Nucleic Acids Res. 37 (Database issue), D674–D679.

Schmidt, E.D., Aguilera, G., Binnekade, R., Tilders, F.J., 2003. Single administration of interleukin-1 increased corticotropin releasing hormone and corticotropin releasing hormone-receptor mRNA in the hypothalamic paraventricular nucleus which paralleled long-lasting (weeks) sensitization to emotional stressors. Neuroscience 116 (1), 275–283.

Shelly, S., Boaz, M., Orbach, H., 2012. Prolactin and autoimmunity. Autoimmun. Rev. 11 (6–7), A465–A470.

Shelton, R.C., Claiborne, J., Sidoryk-Wegrzynowicz, M., Reddy, R., Aschner, M., Lewis, D.A., Mirnics, K., 2011. Altered expression of genes involved in inflammation and apoptosis in frontal cortex in major depression. Mol. Psychiatry 16 (7), 751–762.

Shewale, S.V., Anstadt, M.P., Horenziak, M., Izu, B., Morgan, E.E., Lucot, J.B., Morris, M., 2012. Sarin causes autonomic imbalance and cardiomyopathy: an important issue for military and civilian health. J. Cardiovasc. Pharmacol. 60 (1), 76–87.

Smets, E.M., Garssen, B., Bonke, B., De Haes, J.C., 1995. The Multidimensional Fatigue Inventory (MFI) psychometric qualities of an instrument to assess fatigue. J. Psychosom. Res. 39 (3), 315–325.

Smoot, M., Ono, K., Ruscheinski, J., Wang, P.-L., Ideker, T., 2011. Cytoscape 2.8: new features for data integration and network visualization. Bioinformatics 27 (3), 431–432.

Storey, J.D., 2002. A direct approach to false discovery rates. J. Royal Stat. Soc. 64 (3), 479–498.

Sun, Z., Asmann, Y.W., Kalari, K.R., Bot, B., Eckel-Passow, J.E., Baker, T.R., Carr, J.M., Khrebtukova, I., Luo, S., Zhang, L., Schroth, G.P., Perez, E.A., Thompson, E.A., 2011. Integrated analysis of gene expression, CpG island methylation, and gene copy number in breast cancer cells by deep sequencing. PLoS ONE 6 (2), e17490.

Tani, M., Hayakawa, H., Yasuda, T., Nihira, T., Hattori, N., Mizuno, Y., Mochizuki, H., 2010. Ectopic expression of α-synuclein affects the migration of neural stem cells in mouse subventricular zone. J. Neurochem. 115 (4), 854–863.

Torres-Altoro, M.I., Mathur, B.N., Drerup, J.M., Thomas, R., Lovinger, D.M., O'Callaghan, J.P., Bibb, J.A., 2011. Organophosphates dysregulate dopamine signaling, glutamatergic neurotransmission, and induce neuronal injury markers in striatum. J. Neurochem. 119 (2), 303–313.

Waickman, A.T., Powell, J.D., 2012. mTOR, metabolism, and the regulation of T-cell differentiation and function. Immunol. Rev. 249 (1), 43–58.

Walker, T.L., Vukovic, J., Koudijs, M.M., Blackmore, D.G., Mackay, E.W., Sykes, A.M., Overall, R.W., Hamlin, A.S., Bartlett, P.F., 2012. Prolactin stimulates precursor cells in the adult mouse hippocampus. PLoS ONE 7 (9), e44371.

Walsh, N.P., Gleeson, M., Shephard, R.J., Gleeson, M., Woods, J.A., Bishop, N.C., Fleshner, M., Green, C., Pedersen, B.K., Hoffman-Goetz, L., Rogers, C.J., Northoff, H., Abbasi, A., Simon, P., 2011. Position statement. Part one: immune function and exercise. Exerc. Immunol. Rev. 17, 6–63, Review.

Ware Jr., J.E., Sherbourne, C.D., 1992. The MOS 36-item short-form health survey (SF-36). I. Conceptual framework and item selection. Med. Care. 30 (6), 473–483.

Weidenfeld, J., Siegel, R., Conforti, N., Mizrachi, R., Brenner, T., 1983. Effect of intracerebroventricular injection of nicotinic acetylcholine receptor antibodies on ACTH, corticosterone and prolactin secretion in the male rat. Brain Res. 265 (1), 152–156.

Whistler, T., Fletcher, M.A., Lonergan, W., Zeng, X.R., Lin, J.M., Laperriere, A., Vernon, S.D., Klimas, N.G., 2009. Impaired immune function in Gulf War Illness. BMC Med. Genomics 2, 12.

Wolf, J.M., Rohleder, N., Bierhaus, A., Nawroth, P.P., Kirschbaum, C., 2009. Determinants of the NF-kappaB response to acute psychosocial stress in humans. Brain Behav. Immun. 23 (6), 742–749.

Wolfe, J., Proctor, S.P., Davis, J.D., Borgos, M.S., Friedman, M.J., 1998. Health symptoms reported by Persian Gulf War veterans two years after return. Am. J. Ind. Med. 33 (2), 104–113.

Wu, Z., Irizarry, R.A., 2005. Stochastic models inspired by hybridization theory for short oligonucleotide arrays. Proc. RECOMB 2004. J. Comput. Biol. 12 (6), 882–893.

Wu, Z., Irizarry, R.A., Gentleman, R., Murillo, F.M., Spencer, F., 2004. A Model Based Background Adjustment for Oligonucleotide Expression Arrays. J. Am. Stat. Assoc. 99 (468), 909–917.

Yanagida, K., Ishii, S., Hamano, F., Noguchi, K., Shimizu, T., 2007. LPA4/p2y9/GPR23 mediates rho-dependent morphological changes in a rat neuronal cell line. J. Biol. Chem. 282 (8), 5814–5824.

Yu-Lee, L.Y., 2002. Prolactin modulation of immune and inflammatory responses. Recent Prog. Horm. Res. 57, 435–455.

Zhang, Q., Zhou, X.D., Denny, T., Ottenweller, J.E., Lange, G., LaManca, J.J., Lavietes, M.H., Pollet, C., Gause, W.C., Natelson, B.H., 1999. Changes in immune parameters seen in Gulf War veterans but not in civilians with chronic fatigue syndrome. Clin. Diagn. Lab. Immunol. 6 (1), 6–13.

Zhu, H., O'Brien, J.J., O'Callaghan, J.P., Miller, D.B., Zhang, Q., Rana, M., Tsui, T., Peng, Y., Tomesch, J., Hendrick, J.P., Wennogle, L.P., Snyder, G.L., 2010. Nerve agent exposure elicits site-specific changes in protein phosphorylation in mouse brain. Brain Res. 1342, 11–23.